Dongwoo Kwak

LLM inference engineer: vLLM porting and profiling, Triton kernels, NPU/GPU serving, C++/Python since 2024
Seoul, South Korea · dong0881414@gmail.com · github.com/go-deung-eo · Portfolio

Experience

Dnotitia Seoul, South Korea
Optimization Engineer, AI Group – Inference Optimization *Feb 2026 – Present*

- Ported the team's 2-bit n-gram-table quantization into vLLM for 3 MoE LLMs with Triton gather-decode kernels: on 4×H200, weights 84.7→64.0 GiB/rank, +54% KV cache, TPOT within 0.6%; made DeepSeek-V4.1-Flash servable.
- Cut first-turn TTFT 96% (24.4 s → 0.9 s, 5.7K-token prompt) for an NPU-served Qwen3.6-35B-A3B agent via a session warm-up that pre-fills the byte-identical prefix; owned agent–engine integration in a 3-person PoC.
- Traced MTP's slowdown on a 512-expert MoE (1.54× →0.83×, 1→16 sessions, full acceptance; dense 2.1–2.7×) to route-proportional FFN cost (nsys); a 55k-token routing replay stopped an expert-tiering plan.
- Benchmarked 3 engines incl. vLLM and SGLang for an in-house 2B embedding model under a cosine ≥ 0.999 gate (66,810 tokens/s; 358 img/s, 3.1× SGLang); found API errors behind HTTP 200 that cut MTEB 0.848→0.718.
- Sized KV/attention offload of a 27B dense LLM to an in-house accelerator before hardware existed: designed latency-injection runs (1–64 sessions) and a 256K/1M projection showing 3.5× resident long chats per card.
- Built a bilingual pass@1 agent eval harness (8 models); traced garbled tokens to vLLM's default temperature 1.0.

Klleon Seoul, South Korea
AI Engineer, Product AI Team *Mar 2025 – Jan 2026*

- Fixed a GPU memory leak unresolved for 8 months by tracing missing shared-memory unregistration in Triton client teardown, reaching zero OOM crashes and retiring a forced restart every 3 days.
- Cut TTS request volume 75% and end-to-end latency up to 80% by building dynamic sentence batching on a CUDA-Graph GPT-SoVITS server inside the real-time digital-human engine (C++17).
- Reduced AWS GPU nodes 40% by moving media-engine processes off Triton inference nodes (10:1 → 50:3 ratio).
- Detected broken frames of a lightweight face model in 0.85 ms/frame (OpenCV, CUDA): 2× processes/pod.

Tmax Cloud Seongnam, South Korea
Researcher, Middleware Research Division *Feb 2024 – Oct 2024*

- Built a Triton-based model-serving IaaS (v0.0.0–v0.2.0) in C++20, porting the Triton core C API for asynchronous inference, health and statistics calls and adding tensor binary encoding.
- Sped up JSON writes 7× and reads 3× by replacing boost::property_tree with glaze in the middleware framework.
- Automated a 4-stage GitLab CI/CD pipeline (build, image, E2E test, deploy) with a Python E2E client.

Projects

vLLM Serving Toolkit | *Bash, Python, FastAPI, vLLM* 2026

- Wrote a layered-flag vLLM launcher, logging proxy and smoke harness measuring TTFT, TPOT and tool-call success; compared 7 models on RTX 6000 Ada vs GB10 and reused it in 4 later benchmark campaigns.

vLLM NGQ2 Port (public fork) | *Python, Triton, vLLM 0.31* 2026

- go-deung-eo/vllm (ngq2-ple): 2-bit table loaders, TP sharding, pinned-host placement for 3 models (12 files, +822/−16), bit-exact on a 320M-row table under CUDA Graphs (81 pytest cases).

Skills

Languages: C++17/20, Python, CUDA (working), TypeScript
Inference: vLLM, SGLang, Triton Inference Server, PyTorch/LibTorch, Nsight Systems, OpenCV, prefix/KV caching
Systems: Linux (epoll, sockets), Docker, Kubernetes, GitLab CI/CD, CMake, Meson, Conan, Grafana
Certifications: SQLD (SQL Developer); English OPIc IH (2026)

Education

Yonsei University Seoul, South Korea
B.A. in Economics, Minor in Computer Science; GPA 3.56/4.5 *Mar 2018 – Feb 2024*

- Operating Systems (10/130), Databases, Computer Architecture, Data Mining; 42 Seoul (2021–2023).